

BOLU, TurkeyWMO: **170700**Lat: **40.7338N**Lon: **31.6016E**Elev: **2438**StdP: **13.45**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.2	20.5	12.5	11.4	17.8	16.3	13.7	22.8	9.9	43.2	8.9	40.3	2.3	90	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.3	89.2	67.3	86.0	67.1	83.1	66.1	71.1	83.2	69.4	81.3	67.9	79.5	5.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.1	109.0	78.2	65.4	102.7	75.9	63.9	97.4	74.2	36.6	83.4	35.1	81.4	33.7	79.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.2	7.4	8.6	96.3	5.1	3.2	4.9	98.6	1.9	100.5	-0.9	102.3	-4.6	104.7		
(5)			8.0	74.5	5.0	1.5	4.4	75.5	1.4	76.4	-1.4	77.2	-5.0	78.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.0	35.6	38.1	42.8	50.9	59.4	65.2	69.8	70.5	63.5	55.8	46.0	38.0	(6)
(7)		DBStd	14.00	7.58	8.30	7.82	7.50	6.34	5.21	4.42	4.49	5.54	6.82	7.24	8.09	(7)
(8)		HDD50	1680	449	339	245	76	7	0	0	0	0	26	157	381	(8)
(9)		HDD65	4836	912	753	687	424	189	60	9	10	91	292	571	838	(9)
(10)		CDD50	2792	1	6	23	104	300	455	615	635	404	205	36	8	(10)
(11)		CDD65	473	0	0	0	2	16	65	159	180	45	6	0	0	(11)
(12)		CDH74	4642	0	0	3	53	290	775	1455	1534	447	85	0	0	(12)
(13)		CDH80	1424	0	0	0	6	54	222	484	531	112	15	0	0	(13)
(14)	Wind	WSAvg	2.8	2.4	2.7	3.1	3.1	2.9	3.0	3.1	3.0	2.9	2.3	2.2	2.3	(14)
(15)	Precipitation	PrecAvg	21.7	2.3	1.7	1.9	2.0	2.4	2.1	1.1	1.0	1.1	1.6	1.8	2.7	(15)
(16)		PrecMax	28.2	5.1	4.8	3.9	3.6	6.9	6.1	3.7	3.1	3.1	3.1	4.5	6.1	(16)
(17)		PrecMin	14.8	0.7	0.3	0.4	0.3	0.9	0.4	0.1	0.0	0.0	0.2	0.1	0.6	(17)
(18)		PrecStd	3.2	1.1	1.0	0.9	0.7	1.1	1.2	0.9	0.9	0.8	0.8	1.1	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	64.4	71.0	80.1	85.3	90.3	94.3	93.5	88.1	81.6	70.0	63.7	(19)
(20)			MCWB	46.4	49.3	52.2	58.1	62.4	66.7	68.2	66.3	64.3	62.3	55.0	50.0	(20)
(21)		2%	DB	53.5	59.1	65.6	73.9	80.7	85.8	88.8	89.1	83.0	75.7	64.6	56.7	(21)
(22)			MCWB	45.2	47.8	50.2	56.1	61.3	66.8	68.6	67.6	63.8	59.5	52.3	47.8	(22)
(23)		5%	DB	50.2	54.5	60.9	69.5	77.0	82.0	85.3	85.7	79.3	71.3	60.7	52.9	(23)
(24)			MCWB	43.6	45.4	48.3	54.2	60.6	65.5	68.2	67.1	63.4	58.2	51.2	46.3	(24)
(25)		10%	DB	47.3	50.6	56.5	65.1	73.4	78.5	81.9	82.4	75.3	66.8	56.9	49.6	(25)
(26)			MCWB	41.9	43.8	46.5	52.4	59.4	64.5	67.0	66.6	61.7	57.0	49.3	44.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	51.3	53.9	60.6	66.0	71.2	73.6	73.2	68.8	64.6	56.7	51.5	(27)
(28)			MCD B	55.3	61.4	67.4	74.4	78.9	83.5	86.4	85.1	81.0	76.7	65.9	61.1	(28)
(29)		2%	WB	45.8	48.2	50.9	57.5	63.6	68.6	71.6	70.8	66.3	61.5	54.0	48.6	(29)
(30)			MCD B	52.3	56.7	62.9	71.2	76.4	81.2	83.9	83.0	77.6	71.7	61.7	55.1	(30)
(31)		5%	WB	43.9	46.2	48.9	55.3	61.8	66.7	69.8	69.2	64.5	59.5	52.0	46.8	(31)
(32)			MCD B	49.6	53.8	59.3	67.3	73.8	78.9	81.3	81.3	75.9	68.5	58.9	52.3	(32)
(33)		10%	WB	42.0	44.0	46.9	53.1	60.0	65.0	68.1	67.6	62.9	57.7	50.0	44.6	(33)
(34)			MCD B	46.7	50.5	55.6	63.4	71.0	76.3	79.3	79.1	73.1	65.2	55.9	49.5	(34)
(35)	Mean Daily Temperature Range	MDBR		12.9	15.4	17.7	20.5	22.3	22.8	23.1	23.3	22.6	19.5	17.6	12.9	(35)
(36)		5% DB	MCD BR	18.1	23.1	27.3	29.1	29.6	30.3	29.3	29.8	30.0	27.7	25.2	18.0	(36)
(37)			MCWBR	11.2	13.1	14.1	13.9	13.6	14.3	13.0	12.1	13.1	13.2	13.9	10.8	(37)
(38)		5% WB	MCD BR	16.7	21.0	24.8	25.9	26.2	27.0	25.3	24.2	26.2	24.3	21.2	15.5	(38)
(39)			MCWBR	10.6	12.2	13.2	13.1	13.0	14.0	12.6	11.5	12.3	12.1	12.0	9.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.314	0.340	0.384	0.419	0.397	0.380	0.380	0.381	0.367	0.355	0.325	0.314	(40)
(41)		taud		2.370	2.306	2.218	2.158	2.294	2.375	2.386	2.376	2.412	2.427	2.466	2.429	(41)
(42)		Ebn at Noon		266	274	273	271	279	284	282	279	276	266	261	257	(42)
(43)		Edh at Noon		27	33	41	46	41	38	37	37	33	29	24	23	(43)
(44)	All-Sky Solar Radiation	RadAvg		512	791	1153	1536	1880	2136	2253	1999	1516	1029	703	475	(44)
(45)		RadStd		67	121	110	164	170	201	101	126	121	86	88	63	(45)